COMP 3920

DATABASE SYSTEMS 1

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SUMMARY TABLES

- Useful for Reporting
- Aggregates on large tables (> 1,000,000 rows)
  Group By’s with Count(), Avg(), Sum(), Min(), Max()
- When Indexes aren’t enough

SUMMARY TABLES

Data

DateTime	Employee	Product	Amount
2018-11-05 11:05 AM	Frank	Remote Control Train Set	$1,058.12
2018-11-05 1:05 PM	Frank	Remote Control Car	$298.23
2018-11-05 2:05 PM	Frank	Jigsaw Puzzle	$50.76
2018-11-05 3:43 PM	Frank	Rubics Cube	$20.48
2018-11-05 5:12 PM	Frank	Fullsize Batman Replica	$764.52
2018-11-05 1:45 PM	Sara	Coloring Books	$72.71
2018-11-05 12:06 PM	Sara	Kids Racing Card Bed	$1,208.55
2018-11-06 2:44 PM	James	Board Games	$62.64
2018-11-06 4:21 PM	James	Lego Sets	$232.66
2018-11-06 4:57 PM	James	Animal Costumes	$158.22
2018-11-06 3:52 PM	Sara	Glitter Glue Craft Set	$44.22
2018-11-06 5:32 PM	Sara	Lego Sets	$106.77

SUMMARY TABLES

Summary Data

Date	Employee	Amount	SalesCount
2018-11-05	Frank	$2,192.11	5
2018-11-05	Sara	$1,281.26	2
2018-11-06	James	$453.52	3
2018-11-06	Sara	$150.99	2

SUMMARY TABLE QUERIES

```
SELECT COUNT(*)
FROM table
WHERE
  YEAR(DateTime) = 2018 AND
  MONTH(DateTime) = 11
GROUP BY
  MONTH(DateTime),
  YEAR(DateTime)
```

```
SELECT SUM(SalesCount)
FROM summary_table
WHERE
  YEAR(DateTime) = 2018 AND
  MONTH(DateTime) = 11
GROUP BY
  MONTH(DateTime),
  YEAR(DateTime)
```

Same results, but calculations are faster (based on 4 rows vs 12 rows).

DISASTER EVENTS

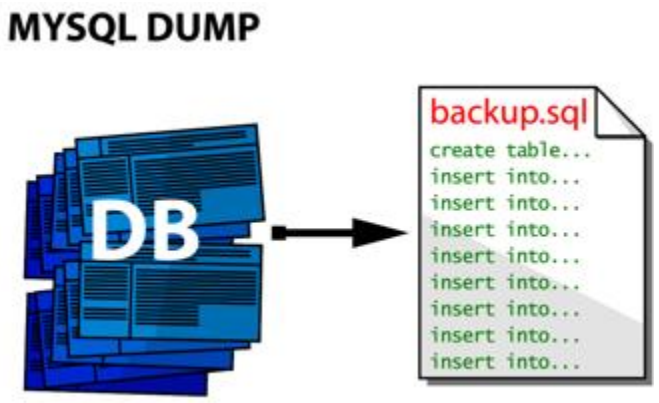
https://www.cloudbacko.com/images/common/diagram_disaster.png

DISASTER EVENTS

- Storage media failure
- Accidental Delete, Drop, Update, Alter Table
- Development bugs / code rollbacks
- Data center natural disaster / theft
- Virus or code bug / corruption
- MySQL database corruption (tables, log files)

LOGICAL VS PHYSICAL

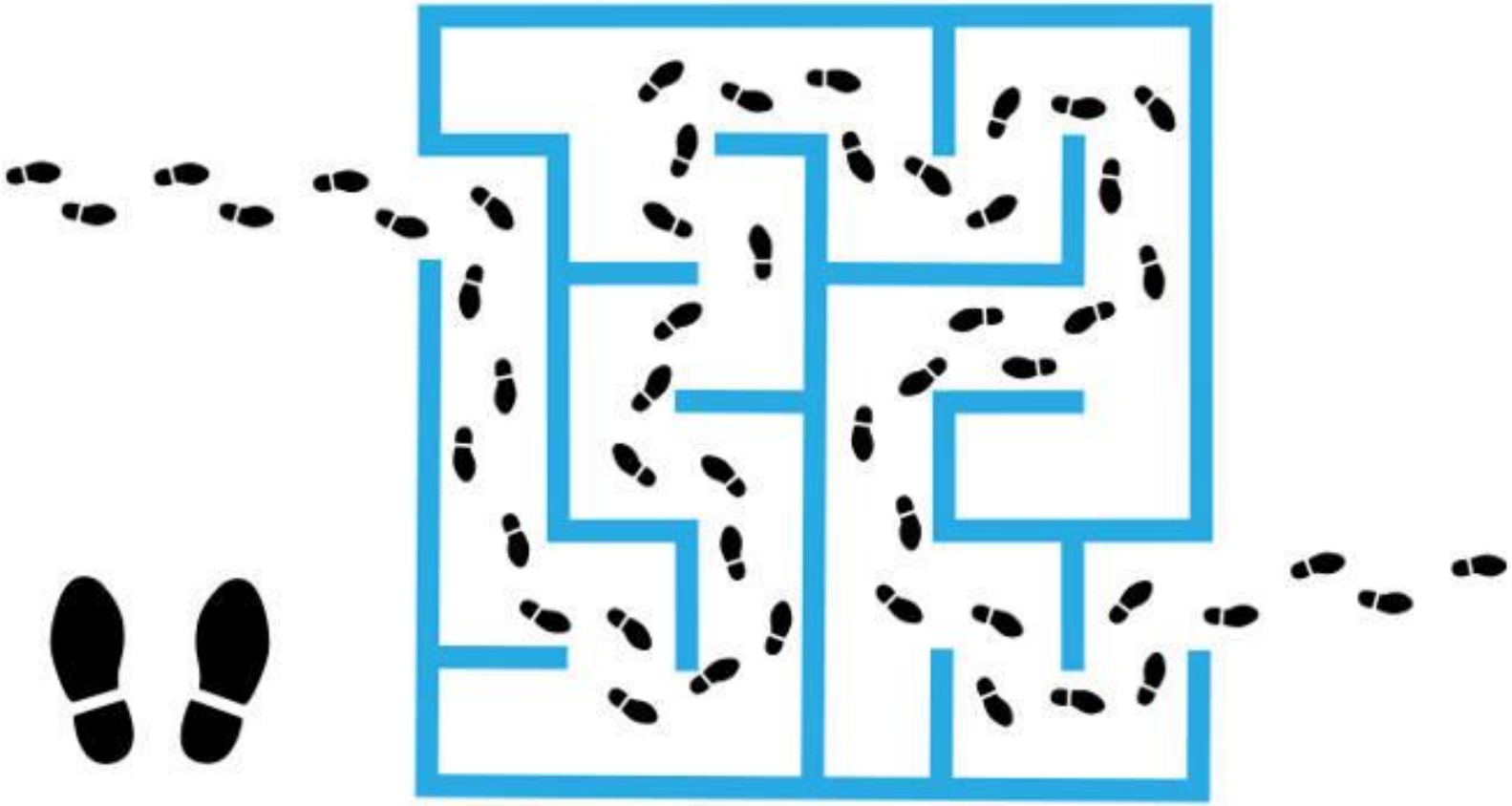
- Logical
  - mysqldump
  - SQL Files
  - Full or Partial for partial restores



- Raw (Physical)
  - XtraBackup or copy of MySQL data directory
  - Binary Files
  - Fast, can recover from corrupt MySQL log files

http://archive.oreilly.com/oreillyschool/courses/dba2/images/07-mysqldump.png
https://cdn3.iconfinder.com/data/icons/folders-and-files-1/60/Folders_and_Files_-_Outline_-_098_-_Binary_File-512.png

LAB WALKTHROUGH



https://theeducatedlandlord.com/wp-content/uploads/2019/08/Tenant-Move-In-Landlord-tenant-walkthrough.jpg